
DELTA: Edge-Centric Dual Attention for Compositional Reasoning on Knowledge Graphs

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Abstract

Graph neural networks treat edges as passive conduits for message passing between nodes, limiting their ability to reason about *relationships between relationships*—the compositional structure underlying multi-hop inference in knowledge graphs. We introduce **DELTA** (Dual Edge-Linked Transformer Architecture), which elevates edges to first-class computational entities via *edge-to-edge attention*: edges attend to structurally adjacent edges through a multi-hop edge adjacency matrix, enabling direct relational composition without node-mediated bottlenecks. On three synthetic structural benchmarks (edge-type relation classification, noise robustness at 80% corruption, and compositional relation discovery), evaluated across 5 seeds on an H100, DELTA outperforms GraphGPS and GRIT by +0.573, +0.290, and +0.095 accuracy respectively—providing the primary cross-task evidence that edge-to-edge attention captures a structurally distinct reasoning capacity. On the complete 14,541-entity FB15k-237 graph (mean degree ≈ 4.1), a 3-seed hop-depth ablation confirms that 2-hop edge adjacency improves 3-hop chain query MRR by +0.012 over 1-hop adjacency (0.218 vs. 0.206), validating the core mechanism at realistic scale. As an exploratory characterization on a top-500-entity degree-based subgraph (mean degree ≈ 19.7), DELTA-Matched (158K parameters) is the only model among 7 tested with a depth-monotonic MRR improvement from 2-hop (0.758) to 5-hop (0.790); GraphGPS degrades from 0.754 to 0.690, and the advantage reaches +0.100 MRR at 5-hop. Cross-seed validation confirms the subgraph trend (0.742 ± 0.009 vs. 0.713 ± 0.007 at 3-hop; non-overlapping CIs). However, a direct mechanism ablation on this dense subgraph finds 1-hop, 2-hop, and node-only conditions statistically indistinguishable, consistent with high link density saturating 1-hop adjacency coverage. An emergent edge–node temperature asymmetry arises during training: edge attention heads converge to sharper distributions while node heads prefer softer averaging, a pattern discovered autonomously via gradient descent. DELTA achieves this with 31% fewer parameters than GraphGPS and faster per-query inference ($0.8\text{--}0.9\times$), at the cost of a one-time encoding overhead from edge adjacency construction. The central open question for future work is whether a SOTA-performing DELTA variant—one incorporating specialized KGC training strategies at full-graph scale—preserves the depth-monotonic compositional advantage observed here.

1 Introduction

Knowledge graph completion (KGC) requires inferring missing links from observed relational structure. While standard link prediction evaluates single-hop tail prediction (“1p”), real-world reasoning often demands *compositional* multi-hop inference: chaining relations such as `born_in`

37 ◦ `located_in` to derive nationality. This compositional capacity is precisely where existing
38 architectures falter.

39 Embedding methods (TransE [1], DistMult [2], RotatE [3]) represent entities and relations as vectors
40 without structural inductive bias. They excel at memorizing local link statistics but collapse on
41 multi-hop chains: DistMult’s MRR drops from 0.728 (2-hop) to 0.457 (5-hop) on our top-500-entity
42 subgraph benchmark. Message-passing GNNs (CompGCN [7], R-GCN [6]) and graph transformers
43 (GraphGPS [10]) propagate information through nodes, treating edges as scalar weights or simple
44 feature carriers. This design creates an information bottleneck: to reason about relationships between
45 relationships, information must route through intermediate nodes, accumulating compression loss at
46 each hop.

47 We propose **DELTA**, an architecture that makes edges first-class computational entities. DELTA’s
48 key mechanism is *edge-to-edge attention via multi-hop edge adjacency*: two edges that share a node
49 are “adjacent” in edge space, and at 2 hops, an edge can attend to edges two steps away. This creates
50 a direct channel for relational composition—`born_in` can directly attend to `located_in` without
51 routing through the intermediate city node—enabling transitive inference as a natural consequence of
52 the attention structure rather than an emergent property that must be learned.

53 DELTA runs node attention and edge attention as parallel streams, reconciled through a bidirectional
54 bridge that ensures cross-stream coherence. This *dual parallel attention* converges $2.7\times$ faster than
55 sequential alternatives while maintaining the same representational capacity.

56 Contributions:

- 57 1. **Edge-to-edge attention architecture.** We introduce dual parallel attention with multi-hop
58 edge adjacency, the first GNN mechanism that enables direct edge-to-edge reasoning without
59 node-mediated bottlenecks (§3).
- 60 2. **Cross-task synthetic validation.** On three synthetic structural benchmarks—edge-type
61 relation classification, edge classification under 80% noise corruption, and compositional re-
62 lation discovery (4 base + 3 derived rules)—DELTA achieves 0.880, 1.000, and 1.000 mean
63 accuracy (5 seeds, H100), vs. GraphGPS 0.293/0.697/0.905 and GRIT 0.307/0.710/0.893.
64 These tasks are entirely distinct from knowledge graph completion and provide cross-task
65 evidence that edge-to-edge attention captures a general structural reasoning capacity (§4.2).
- 66 3. **Full-scale validation on complete FB15k-237.** A direct 3-seed hop-depth ablation on the
67 full 14,541-entity graph confirms that 2-hop edge adjacency improves 3-hop chain query
68 MRR by +0.012 over 1-hop (0.218 vs. 0.206) on the sparse graph (mean degree ≈ 4.1)
69 where the mechanism is most relevant (§4.5).
- 70 4. **Exploratory depth characterization on degree-based subgraph.** On a 500-entity degree-
71 based subgraph of FB15k-237, DELTA-Matched exhibits a depth-monotonic MRR im-
72 provement from 2-hop to 5-hop that no other tested model shares, with the advantage over
73 GraphGPS accelerating from +0.004 to +0.100 MRR. A direct mechanism ablation on
74 this subgraph finds all hop-depth conditions statistically indistinguishable (consistent with
75 high density saturating 1-hop adjacency), so these results characterize emergent behavioral
76 depth-dependence rather than isolating the mechanism (§4.3).
- 77 5. **Emergent attention temperature asymmetry.** We observe that edge and node attention
78 heads develop fundamentally different sharpness preferences during training: edge heads
79 drift toward higher temperature (sharper) while node heads drift lower (softer). This
80 asymmetry is consistent across 20+ configurations and 9 experimental phases, suggesting
81 distinct computational roles for the two attention streams (Appendix I).
- 82 6. **Scalability characterization.** We evaluate DELTA at scales from $N=500$ to $N=14,541$
83 entities, identify attention dilution as the primary scaling bottleneck, and validate top- k
84 sparse attention ($k=128$) as a lossless approximation that preserves full-softmax quality
85 (Appendix H).

86 2 Related Work

87 **Knowledge graph embeddings.** TransE [1] models relations as translations in embedding space;
88 DistMult [2] uses bilinear scoring; RotatE [3] models relations as rotations in complex space;

89 ComplEx [4] extends to complex-valued embeddings. These methods achieve strong single-hop link
 90 prediction but lack structural inductive bias for multi-hop reasoning. PairRE [5] and RotatE achieve
 91 MRR ~ 0.34 on full FB15k-237, but rely entirely on embedding geometry rather than graph structure
 92 for relational composition.

93 **Message-passing GNNs for KGs.** R-GCN [6] introduces relation-specific weight matrices for
 94 typed edges. CompGCN [7] jointly embeds nodes and relations through composition operations
 95 (subtraction, multiplication, circular correlation). Both treat edges as typed message channels
 96 rather than computational entities—edge representations are derived from relation type embeddings,
 97 not learned through edge-to-edge interaction. NBFNet [8] and RED-GNN [9] address multi-hop
 98 reasoning via path-based message passing but still route all information through nodes. A*Net [21]
 99 scales path-based reasoning to large KGs via priority-based entity selection. These path-based
 100 methods are a key baseline comparison target for future work; our primary comparison here is against
 101 general graph transformer architectures (GraphGPS, GRIT).

102 **Graph transformers.** GraphGPS [10] combines local MPNN message passing with global trans-
 103 former attention, using positional/structural encodings to inject graph awareness into the attention
 104 mechanism. GRIT [11] uses random-walk structural encodings with graph transformers. Both archi-
 105 tectures provide expressive node-level representations but lack mechanisms for direct edge-to-edge
 106 interaction. Expformer [12] addresses scalability via sparse attention patterns but does not introduce
 107 edge-level attention.

108 **Edge-aware architectures.** Line graphs [13] and their GNN variants operate on the dual graph
 109 where edges become nodes, enabling edge-level message passing. However, line graph construction
 110 is $O(E \cdot d_{\max})$ and the resulting graph can be much larger than the original. DELTA’s edge adja-
 111 cency matrix provides similar connectivity at $O(E^{0.97})$ via sparse COO operations, with the crucial
 112 difference that DELTA maintains both node and edge representations simultaneously through dual
 113 attention. EHGNN [14] incorporates edge features into heterogeneous GNNs but does not implement
 114 edge-to-edge attention.

115 **Multi-hop reasoning on KGs.** BetaE [15] and ConE [16] embed logical queries as geometric
 116 regions for answering complex queries. GQE [17] and Q2B [18] project queries into embedding
 117 spaces. CQD [22] answers complex queries by continuous optimization over link predictor scores.
 118 GNN-QE [23] combines GNN encoders with fuzzy logic for complex query answering. These
 119 methods are designed specifically for complex query answering rather than general KGC. Our multi-
 120 hop evaluation uses chain queries (2p–5p) with soft entity traversal—a complementary protocol
 121 that tests compositional capacity within standard KGC training, and is not directly comparable to
 122 BetaE/GQE query benchmark numbers.

123 3 Architecture

124 DELTA processes a knowledge graph $G = (V, E, R)$ with entity set V , edge set $E \subseteq V \times R \times V$,
 125 and relation set R . Each entity v_i has a learned embedding $\mathbf{h}_i \in \mathbb{R}^{d_n}$ and each edge (v_i, r_k, v_j) has a
 126 learned embedding $\mathbf{e}_{ij} \in \mathbb{R}^{d_e}$.

127 3.1 Multi-Hop Edge Adjacency

128 The foundation of DELTA is the *edge adjacency matrix* $\mathbf{A}_E$, which encodes structural connectivity
 129 between edges. Two edges $e_a = (v_i, r, v_j)$ and $e_b = (v_k, r', v_l)$ are 1-hop adjacent if they share an
 130 endpoint: $\{v_i, v_j\} \cap \{v_k, v_l\} \neq \emptyset$. The 2-hop edge adjacency extends this via the incidence structure:

$$\mathbf{A}_E^{(2)} = (\mathbf{B}^\top \mathbf{B})^2 - \text{diag} \quad (1)$$

131 where $\mathbf{B} \in \{0, 1\}^{|V| \times |E|}$ is the node-edge incidence matrix. This is computed efficiently using
 132 sparse COO tensor operations via `torch_sparse.spspmm`, achieving $O(E^{0.97})$ scaling (validated
 133 empirically from 50 to 2,500 edges). At 2 hops, an edge can “see” edges two steps away, enabling
 134 direct reasoning about relational chains such as `born_in` $\rightarrow$ `located_in` without intermediate node
 135 representations.

136 We validate the role of 2-hop edge adjacency in §4.2: on a synthetic transitive relation benchmark,
 137 2-hop achieves 100% accuracy on derived relations vs. 61.1% for 1-hop. On FB15k-237 multi-hop
 138 queries, a direct ablation finds all three adjacency configurations perform comparably (Table 5); we
 139 discuss this result and its implications in §4.5.

140 3.2 Edge-to-Edge Attention

141 Given the edge adjacency $\mathbf{A}_E^{(2)}$ stored as a sparse index $[2, E_{\text{adj}}]$, we compute multi-head edge
 142 attention. For edge e_i attending to adjacent edge e_j :

$$\alpha_{ij}^{(h)} = \frac{\tau^{(h)} \cdot (\mathbf{W}_Q^{(h)} \mathbf{e}_i)^\top (\mathbf{W}_K^{(h)} \mathbf{e}_j)}{\sqrt{d_h}} + \mathbf{W}_{\text{ctx}} [\mathbf{h}_{s(j)} \parallel \mathbf{h}_{t(j)}]_h \quad (2)$$

143 where $\tau^{(h)} = \exp(\theta^{(h)})$ is a learnable per-head temperature (always positive). Note that $\tau^{(h)}$
 144 *multiplies* the dot product (not divides): larger τ produces sharper, more selective attention—the
 145 inverse of the conventional divide-by-temperature softmax formulation. $s(j)$ and $t(j)$ are the source
 146 and target nodes of edge e_j , $\parallel$ denotes concatenation, and $d_h = d_e/H$ is the head dimension. The
 147 node context bias $\mathbf{W}_{\text{ctx}} \in \mathbb{R}^{2d_n \times H}$ grounds edge attention in the endpoint identities of the attending
 148 edges.

149 Attention weights are computed via scatter-softmax over the sparse adjacency structure (normalizing
 150 per target edge), followed by weighted aggregation:

$$\mathbf{e}'_i = \text{LayerNorm} \left(\mathbf{e}_i + \mathbf{W}_O \sum_{j \in \mathcal{N}_E(i)} \text{softmax}(\alpha_{ij}) \cdot \mathbf{W}_V \mathbf{e}_j \right) \quad (3)$$

151 **Top- k sparse attention.** For large graphs where $|\mathcal{N}_E(i)|$ becomes prohibitive, we retain only the k
 152 highest-scoring neighbors per target edge (based on mean attention across heads), then renormalize.
 153 At $N=5,000$ entities with 63M adjacency pairs, $k=128$ preserves full-softmax quality exactly (test
 154 MRR 0.2472 vs. 0.2457 for full softmax), while $k=64$ degrades by -5.6% .

155 3.3 Node Attention

156 Node attention follows a GAT-style [20] design where each node attends to its neighbors, with edge
 157 features providing a per-head bias:

$$\beta_{ij}^{(h)} = \frac{\tau_n^{(h)} \cdot (\mathbf{W}_Q^{(h)} \mathbf{h}_i)^\top (\mathbf{W}_K^{(h)} \mathbf{h}_j)}{\sqrt{d_h}} + \mathbf{W}_{\text{edge}} (\mathbf{e}_{ij})_h \quad (4)$$

158 Node attention uses independent learnable temperatures $\tau_n^{(h)}$ per head, enabling the model to discover
 159 optimal sharpness for each attention stream independently.

160 3.4 Dual Parallel Attention

161 Both attention mechanisms operate on the *same* input state simultaneously:

$$\mathbf{H}' = \text{NodeAttn}(\mathbf{H}, \mathbf{E}, G), \quad \mathbf{E}' = \text{EdgeAttn}(\mathbf{E}, \mathbf{H}, \mathbf{A}_E) \quad (5)$$

162 This parallel design ensures that neither stream’s output biases the other’s computation within a layer,
 163 and converges $2.7\times$ faster than sequential alternatives.

164 3.5 Reconciliation Bridge

165 After parallel attention, a bidirectional bridge ensures cross-stream coherence:

$$\hat{\mathbf{e}}_{ij} = \text{LN}(\mathbf{e}'_{ij} + \mathbf{W}_1[\mathbf{e}'_{ij} \parallel \mathbf{h}'_{s(ij)} \parallel \mathbf{h}'_{t(ij)}]) \quad (\text{edges absorb node context}) \quad (6)$$

$$\hat{\mathbf{h}}_i = \text{LN}(\mathbf{h}'_i + \mathbf{W}_2[\mathbf{h}'_i \parallel \bar{\mathbf{e}}_i]) \quad (\text{nodes absorb edge context}) \quad (7)$$

where $\bar{\mathbf{e}}_i = \text{mean}(\{\hat{\mathbf{e}}_{ij} : j \in \mathcal{N}(i)\})$ aggregates incident updated edge features. Both updates use residual connections with LayerNorm. Edges update first (absorbing node context), then nodes update (absorbing the already node-informed edge features), creating a sequential cascade within the bridge.

3.6 Link Prediction Head

For link prediction, we use a DistMult [2] decoder:

$$s(v_i, r_k, v_j) = \mathbf{h}_i^\top \text{diag}(\mathbf{r}_k) \mathbf{h}_j \quad (8)$$

where $\mathbf{r}_k$ is a learned relation embedding. Training uses 1-vs-all binary cross-entropy with label smoothing (0.1), following standard KGC practice.

3.7 Multi-Hop Scoring

For n -hop path queries $(v_0, r_1, ?, r_2, ?, \dots, r_n, ?)$, we use *soft entity traversal*: at each intermediate hop, score all entities via the DistMult decoder, apply softmax with temperature $\tau = 1.0$ to obtain a probability distribution over entities, take the weighted average entity embedding, and pass it to the next hop. This provides a differentiable approximation to hard entity traversal while remaining faithful to the compositional structure of the query.

4 Experiments

4.1 Setup

Dataset. We use FB15k-237 [19], the standard filtered subset of Freebase with 14,541 entities, 237 relations, and 310,116 triples. Our compositional reasoning and mechanism analysis experiments (§4.3) use a **top-500 degree subgraph** (494 entities, 207 relations, $\sim 9,700$ train triples, mean degree ≈ 19.7) with full filtered evaluation. This degree-biased subgraph enables rapid multi-configuration ablations across many seeds and depth levels. Scaling experiments (Appendix H) extend to $N=2,000$ and $N=5,000$ entity subgraphs. The hop-depth ablation (§4.5) additionally reports a full-scale evaluation on the complete 14,541-entity graph (mean degree ≈ 4.1 , 310,116 triples) to validate key findings at standard benchmark scale.

Models. We compare seven configurations:

- **DELTA-Matched** (158K params): 1 DELTA layer, $d_n=64$, $d_e=32$, 4 heads. Our primary model, designed for parameter-fair comparison.
- **DELTA-Full** (293K params): 3 DELTA layers.
- **SelfBootstrapHybrid** (381K params): A FixedChain DELTA layer bootstraps graph construction before 3 DELTA processing layers.
- **GraphGPS** [10] (228K params): Local MPNN + global transformer attention.
- **GRIT** [11] (197K params): Random-walk structural encodings + graph transformer.
- **DistMult** [2] (47K params): Pure embedding baseline (no GNN).
- **SelfBootstrap** (299K params): DELTA bootstraps its own graph construction.

Training. All models use Adam optimizer ($\text{lr} = 0.003$), batch size 4,096, 1-vs-all BCE loss with label smoothing 0.1, and early stopping with patience 10. Evaluation uses standard filtered MRR and Hits@ k metrics. Multi-hop queries use 10% DropEdge for DELTA (validated in §4.3).

Table 1: Phase 11 — transitive relation classification accuracy by adjacency configuration (100-entity synthetic graph, single seed). Only 2-hop edge adjacency achieves perfect accuracy on derived (composed) relation instances.

Config	Transitive Acc.	Δ vs. 2-hop
2-hop (DELTA)	1.000	—
Node-GNN (no edge attn)	0.833	−0.167
1-hop	0.611	−0.389
MLP (no structure)	0.375	−0.625

Multi-hop query generation. Chain queries are generated by composing training edges to reach test edges: a 2-hop query chains one train edge to one test edge, a 3-hop query chains two train edges to one test edge, etc. All queries undergo strict leakage prevention: no 1-hop shortcuts between anchor and answer in the training graph, no duplicate queries, and automated audit verification before evaluation. Query counts: 1p=486, 2p=5,382, 3p=10,000, 4p=10,000, 5p=10,000.

4.2 Synthetic Structural Benchmarks

Before evaluating on the FB15k-237 knowledge graph, we validate the edge-to-edge attention mechanism on synthetic structural tasks that isolate compositional reasoning without confounds from embedding quality or training scale.

Transitive edge adjacency (Phase 11). On a 100-entity synthetic knowledge graph with explicitly transitive relations, we ablate edge adjacency depth (Table 1). 2-hop edge adjacency achieves **100%** accuracy on derived (transitive) relation instances, vs. 1-hop adjacency at 61.1%, a node GNN at 83.3%, and an MLP at 37.5%. 2-hop adjacency is necessary and sufficient for transitive relational composition. This result, alongside the full-graph ablation (+0.012 on 3p MRR), constitutes the primary mechanistic evidence for the hop-depth claim, since the dense-subgraph mechanism ablation (Phase 66, Table 5) finds all conditions statistically indistinguishable.

Three-task benchmark (Phase 34). We compare DELTA-Matched, GraphGPS, and GRIT on three synthetic structural tasks (5 seeds $\times$ 500 epochs, H100; Table 2):

- **Edge-type classification:** A synthetic KG with 6 relation patterns (8 instances each); classify each edge by its relation type. DELTA achieves 0.880 vs. GraphGPS 0.293 and GRIT 0.307 (advantage +0.573). Edge-to-edge attention learns discriminative edge-type representations that node-centric message passing cannot extract—standard graph transformers have no native edge-classification head, and their node representations lose the information needed to distinguish edge types.
- **Noise robustness:** Repeat edge-type classification with 80% of edges randomly corrupted. DELTA maintains 1.000 accuracy while GraphGPS degrades to 0.697 and GRIT to 0.710. DELTA’s edge-level representation is structurally grounded and robust to corrupted neighborhood signals.
- **Compositional relation discovery:** A KG with 4 base relation types and 3 derived (compositional) rules; models must classify edges including derived instances produced by relational composition. DELTA achieves 1.000 vs. GraphGPS 0.905 (+0.095), directly demonstrating that 2-hop edge adjacency enables compositional rule discovery without supervision.

The cross-task pattern is consistent: DELTA’s largest advantage is on the edge-type classification task (+0.573), where node-centric architectures are structurally mismatched. The advantage is smallest on path composition (+0.095), where multi-hop node-level message passing provides a partial substitute. Together with the transitive adjacency ablation, these results show that DELTA’s mechanism is not FB15k-237-specific—edge-to-edge attention provides a general structural advantage wherever edge identity matters.

Probe architecture note. GraphGPS/GRIT have no native edge-classification head; their probes use mean-pooled endpoint node embeddings, while DELTA’s probe operates on learned edge representations. This asymmetry may inflate DELTA’s advantage on edge-type classification. The path

Table 2: Synthetic structural benchmarks (Phase 34): mean accuracy across 5 seeds (500 epochs, H100), compared at matched parameter budgets. Advantage row shows DELTA vs. the stronger of GraphGPS/GRIT.

Model	Params	Edge Cls.	Noise @0.8	Path Comp.
DELTA-Matched	158K	0.880	1.000	1.000
GraphGPS	228K	0.293	0.697	0.905
GRIT	197K	0.307	0.710	0.893
<i>DELTA advantage</i>	—	+0.573	+0.290	+0.095

Table 3: Multi-hop MRR by reasoning depth (single seed, 0% DropEdge). DELTA-Matched is the only model with higher MRR at 5-hop than 2-hop (net +0.032). All baselines show net degradation from 2-hop to 5-hop.

Model	Params	1p	2p	3p	4p	5p
DELTA-Matched	158K	0.541	0.758	0.753	0.767	0.790
GraphGPS	228K	0.523	0.754	0.727	0.701	0.690
DistMult	47K	0.494	0.728	0.583	0.511	0.457
<i>DELTA – GraphGPS gap</i>		+0.018	+0.004	+0.026	+0.066	+0.100

composition result (+0.095), where all models share the same endpoint prediction protocol, provides a cleaner cross-architecture comparison.

4.3 Compositional Reasoning

We characterize depth-dependent performance on the FB15k-237 degree-based subgraph. DELTA-Matched is the only model among 7 tested that shows a depth-monotonic MRR improvement from 2-hop to 5-hop; GraphGPS and all other baselines degrade. Readers should note that a direct mechanism ablation on the same subgraph (Table 5) finds 1-hop, 2-hop, and node-only conditions statistically indistinguishable—the depth-monotonic pattern is a behavioral property of the model on this graph, not an isolated consequence of the hop-depth setting. The full-graph ablation (§4.5) provides the mechanistic evidence that 2-hop adjacency adds discriminative value on sparse graphs.

The depth advantage accelerates. Table 3 presents the central result. DELTA-Matched’s lead over GraphGPS grows from +0.004 (2p) to +0.100 (5p)—a $25\times$ amplification. Critically, DELTA’s 5p MRR (0.790) *exceeds* its own 2p MRR (0.758). No other model exhibits this property: GraphGPS loses -0.064 MRR from 2p→5p, while DistMult loses -0.271 .

Consistent multi-seed advantage. Table 4 confirms statistical robustness (3 seeds, 10% DropEdge): DELTA improves from 2p (0.730 ± 0.011) to 3p (0.742 ± 0.009), while GraphGPS degrades ($0.727 \rightarrow 0.713$); confidence intervals do not overlap. The full 7-model single-seed comparison (Appendix E, Table 10) shows DELTA-Matched is the only model with a positive 2p→3p delta (+0.005); every other model degrades.

Robustness to regularization. We sweep DropEdge rates (0%–40%) for both DELTA-Matched and GraphGPS (Appendix B). DELTA leads GraphGPS on 3p MRR at all five rates tested, with the advantage ranging from +0.001 (at 20%, within noise) to +0.029 (at 0%). The advantage is statistically meaningful at four of the five rates; the multi-hop advantage is structural, not an artifact of a particular hyperparameter configuration.

4.4 Standard Link Prediction

On standard 1p link prediction (top-500 subgraph), GraphGPS leads DELTA-Matched by 0.018 MRR (0.513 vs. 0.495). Full per-model results in Appendix E (Table 11).

Table 4: Multi-seed 3-hop MRR ($\pm$ std, 3 seeds). Non-overlapping confidence intervals.

Config	Params	1p	2p	3p	2p $\rightarrow$ 3p
DELTA @10% DE	158K	0.543 $\pm$ 0.006	0.730 $\pm$ 0.011	0.742 $\pm$ 0.009	+0.012
GraphGPS	228K	0.529 $\pm$ 0.009	0.727 $\pm$ 0.007	0.713 $\pm$ 0.007	-0.014

Table 5: Hop-depth ablation (Phase 66): DELTA-Matched on FB15k-237 top-500 subgraph, 3 seeds $\times$ 500 epochs. Values are mean $\pm$ std. All differences are within 1σ ; no condition is statistically superior.

Config	LP MRR	1p MRR	2p MRR	3p MRR	2p $\rightarrow$ 3p
2-hop (DELTA)	0.496 $\pm$ 0.003	0.534 $\pm$ 0.008	0.726 $\pm$ 0.014	0.731 $\pm$ 0.006	+0.005
1-hop	0.498 $\pm$ 0.008	0.551 $\pm$ 0.001	0.721 $\pm$ 0.007	0.729 $\pm$ 0.007	+0.008
Node-only	0.505 $\pm$ 0.006	0.553 $\pm$ 0.006	0.728 $\pm$ 0.011	0.742 $\pm$ 0.012	+0.014

4.5 Ablation: 1-Hop vs. 2-Hop Edge Adjacency

To directly validate the role of 2-hop edge adjacency—the architectural mechanism underlying DELTA’s compositional reasoning—we ablate three configurations on the FB15k-237 subgraph using the multi-hop evaluation protocol:

- **1-hop**: edges attend only to directly adjacent edges (share a node)
- **2-hop**: edges attend to edges two steps away (default DELTA)
- **Node-only**: edge attention stream disabled; DELTA reduces to GAT-style node attention

Table 5 shows all three configurations are statistically indistinguishable on the dense subgraph (every pairwise gap within one standard deviation). The node-only condition achieves the highest point estimates, though within noise—consistent with 1-hop adjacency already saturating the structural neighborhood at mean degree ≈ 19.7 . This motivates the full-graph ablation below, where 1-hop neighborhoods are genuinely sparse.

Full-Scale Ablation (Phases 67–68). On the complete 14,541-entity graph (mean degree ≈ 4.1 , top- $k=128$ sparse attention, 1,064K parameters, 3 seeds), hops=2 achieves **+0.012** on 3p MRR over hops=1, exceeding the pre-specified +0.010 threshold. (Preliminary 2-seed results showed +0.017; seed 456’s hops=2 3p score revised the 3-seed average to +0.012, still above threshold.) LP MRR is statistically equivalent (0.204 vs. 0.205). On a random 2,242-entity subgraph (uniformly sampled, mean degree 7.6, equal adjacency budget), the 3p gap is +0.010 over 3 seeds. Full multi-metric breakdowns in Appendix F (Table 12 Phase 67 and Table 13 Phase 68). Table 6 synthesizes all three density regimes.

Table 6: Cross-density summary: hops=2 vs. hops=1 on 3p MRR (Phases 66–68). The gap decreases monotonically with mean degree. Absolute 3p values are not directly comparable across rows.

Phase	Subgraph	Mean Deg	hops=1 3p	hops=2 3p	Δ 3p
67	Full FB15k-237 ($N=14,541$)	4.1	0.206 $\pm$ 0.005	0.218 $\pm$ 0.008	+0.012
68	Random ($N=2,242$)	7.6	0.274 $\pm$ 0.060	0.284 $\pm$ 0.031	+0.010
66	Top-degree ($N=500$)	19.7	0.729 $\pm$ 0.007	0.731 $\pm$ 0.006	+0.002

Across all three density regimes the hops=2 advantage decreases monotonically with mean degree, consistent with 1-hop adjacency becoming informationally sufficient above a density threshold.

For inference timing, scaling analysis, and attention temperature mechanistic details, see Appendix G–I.

5 Discussion

Why does the advantage grow with depth? We hypothesize that 2-hop edge adjacency creates a multiplicative compositional advantage: DELTA composes relational information directly in edge space, while node-based architectures decompress through node representations at each hop. At 1 hop this overhead is negligible; at 5 hops, accumulated compression loss in node-mediated architectures becomes substantial. The temperature asymmetry (Appendix I)—edge attention sharper than node attention—supports this interpretation: the model autonomously discovers differential sharpness that mirrors the two streams’ distinct computational roles.

When does the mechanism matter? The Phase 66 dense-subgraph ablation (mean degree ≈ 19.7) finds all adjacency conditions statistically indistinguishable, while the Phase 67 full-graph ablation (mean degree ≈ 4.1) finds $+0.012$ on 3p MRR. The mechanism matters when graph sparsity creates relational chains that 1-hop adjacency cannot bridge. Table 6 confirms this monotonically across all three density regimes.

Does the compositional inductive bias trade off against 1p performance? DELTA-Matched (158K parameters) consistently outperforms DELTA-Full (293K) on multi-hop tasks, which we interpret as larger models memorizing local link statistics at the expense of compositional generalization. The full-graph LP gap below SOTA (0.205 vs. 0.338–0.415) likely reflects training-strategy differences—self-adversarial negative sampling, longer schedules, entity-specific regularization—rather than an architectural incompatibility. The per-scale gap on the top-500 subgraph is only -0.018 against GraphGPS (97% of its MRR at 69% of parameters), suggesting the mechanism itself does not preclude competitive 1p performance; optimizing for it is an engineering question, not a structural limitation.

6 Conclusion

We have presented DELTA, an edge-centric architecture using edge-to-edge attention over multi-hop adjacency. A consistent picture emerges across three experimental contexts: edge-to-edge attention provides a structural reasoning advantage that grows with compositional depth and matters most when graph sparsity creates relational chains that 1-hop adjacency cannot bridge. On synthetic benchmarks, DELTA outperforms GraphGPS and GRIT by $+0.573$, $+0.290$, $+0.095$ (5 seeds, cross-task). On complete FB15k-237 (3 seeds), 2-hop adjacency yields $+0.012$ on 3p MRR over 1-hop, confirming the mechanism at realistic scale. On the degree-based subgraph (7 models, 5 depths), DELTA-Matched is the only model with depth-monotonic MRR improvement (GraphGPS: -0.064), while a direct ablation on the same dense subgraph finds conditions statistically indistinguishable—consistent with the mechanism mattering most when 1-hop adjacency cannot saturate the structural neighborhood. An emergent edge–node temperature asymmetry provides mechanistic interpretability.

These results suggest that the standard single-hop link prediction benchmark undervalues architectures with strong relational inductive biases. Multi-hop evaluation provides a more discriminating test of compositional capacity, and we encourage its adoption alongside standard metrics. We note clearly that DELTA’s full-graph LP MRR (≈ 0.205) falls below published SOTA methods (≈ 0.34 – 0.41), reflecting the gap between demonstrating a novel mechanism and achieving optimized end-to-end performance—a gap that remains an important open problem.

Future work includes (1) full-depth evaluation (2-hop→5-hop) on complete FB15k-237, comparison against NBFNet [8] and A*Net [21], and BetaE/GQE benchmarks [15]; and (2) cross-dataset validation (YAGO3-10, WN18RR, ogbl-wikikg2) and closing the SOTA LP gap via specialized KGC training while preserving the compositional advantage.

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391 A Hits@10 by Depth

Table 7: Hits@10 across reasoning depths (single seed, same experiment as Table 3).

Model	1p	2p	3p	4p	5p
DELTA-Matched	0.854	0.889	0.881	0.885	0.895
GraphGPS	0.848	0.897	0.859	0.818	0.826
DistMult	0.800	0.858	0.744	0.672	0.587

392 B DropEdge Robustness

Table 8: 3-hop MRR across DropEdge rates. DELTA-Matched leads at all rates.

Model	0%	10%	20%	30%	40%
DELTA-Matched	0.740	0.744	0.724	0.732	0.744
GraphGPS	0.711	0.716	0.723	0.720	0.725
Δ	+0.029	+0.028	+0.001	+0.012	+0.019

393 C Brain Architecture (Preliminary)

394 The BrainEncoder module uses Gumbel-sigmoid differentiable edge selection to construct new edges
 395 from learned node representations, then processes the augmented graph (original + constructed edges)
 396 through DELTA layers. On the FB15k-237 subgraph ($N=494$):

Table 9: BrainEncoder results (multi-seed, density $d=0.01$, 200 epochs).

Config	MRR	H@1	H@10	Constructed edges
DELTA-Full	0.494	0.355	0.760	0
brain_hybrid ($d=0.01$)	0.484 ± 0.010	0.334 ± 0.011	0.799 ± 0.006	2,435

397 The brain_hybrid configuration matches DELTA-Full on MRR while delivering +4.7% Hits@10,
 398 demonstrating that differentiable graph construction adds genuine recall through learned structural
 399 augmentation. Density is a critical hyperparameter: $d=0.01$ (2,435 edges) strictly dominates $d=0.02$
 400 (4,870 edges) on all metrics.

401 D Domain Transfer (Preliminary)

402 A frozen DELTA encoder trained on FB15k-237, evaluated via a linear probe on WN18RR, achieves
 403 accuracy of **0.961**—without any domain adaptation mechanism (no GRL, no fine-tuning). **Note:** this
 404 evaluation uses 100 training samples, giving approximate $\pm 5\%$ confidence intervals; results should
 405 be treated as a preliminary signal, not a statistically validated finding. A proper transfer evaluation
 406 requires $\geq 10,000$ samples with confidence intervals on held-out test splits. We report this result as a
 407 directional indicator that DELTA’s learned edge representations may capture transferable relational
 408 structure.

409 E Full Multi-Hop and Link Prediction Results

410 **Standard link prediction (1p).** GraphGPS leads on 1p MRR (0.513 vs. DELTA-Matched 0.495,
 411 -0.018 gap), while DELTA-Matched achieves 97% of GraphGPS’s MRR at 69% of parameters (158K
 412 vs. 228K). All subgraph models substantially exceed published full FB15k-237 results (DistMult
 413 0.241, CompGCN 0.355), reflecting the higher link density of the degree-biased subgraph.

Table 10: Complete multi-hop results (7 models, seed=1, 0% DropEdge). Only DELTA-Matched improves from 2p→3p.

Model	Params	1p	2p	3p	3p H@10	2p→3p
DELTA-Matched	158K	0.533	0.733	0.738	0.873	+0.005
SBHybrid	381K	0.541	0.722	0.695	0.833	−0.027
GraphGPS	228K	0.549	0.718	0.697	0.850	−0.021
DELTA-Full	293K	0.524	0.711	0.692	0.836	−0.020
SelfBootstrap	299K	0.512	0.712	0.686	0.829	−0.026
GRIT	197K	0.460	0.712	0.644	0.783	−0.068
DistMult	47K	0.532	0.715	0.566	0.710	−0.150

Table 11: Standard link prediction (1p) on FB15k-237 top-500 subgraph. Best-validation checkpoint, seed=1.

Model	Params	MRR	H@1	H@3	H@10	Peak Ep
GraphGPS	228K	0.513	0.375	0.581	0.813	200
SBHybrid	381K	0.509	0.363	0.587	0.816	250
DELTA-Matched	158K	0.495	0.346	0.572	0.804	200
DELTA-Full	293K	0.494	0.355	0.559	0.792	200
DistMult	47K	0.484	0.346	0.548	0.763	500+
GRIT	197K	0.439	0.295	0.496	0.760	200

414 F Full Ablation Results (Phases 67–68)

Table 12: Phase 67 — Full FB15k-237 hop-depth ablation (3 seeds × 200 epochs, top- k =128 sparse attention, 1,064K parameters). Values are mean ± std across all 3 seeds. 2p/3p queries use 10,000 samples each.

Config	LP MRR	H@10	1p MRR	2p MRR	3p MRR	2p→3p [†]
hops=1	0.205 ± 0.002	0.381 ± 0.003	0.257 ± 0.004	0.113 ± 0.003	0.206 ± 0.005	+0.093
hops=2	0.204 ± 0.002	0.382 ± 0.002	0.252 ± 0.003	0.116 ± 0.003	0.218 ± 0.008	+0.102
Δ (2−1)	−0.001	+0.001	−0.005	+0.003	+0.012	—

[†]Absolute 2p→3p improvement (3p MRR − 2p MRR). Both conditions show positive improvement from 2p to 3p on the full graph.

415 Per-seed variance for Phase 67 is low: hops=1 seeds span 3p MRR 0.203–0.214; hops=2 seeds
 416 42 and 123 exceed their hops=1 counterparts (3p 0.219–0.227 vs. 0.203–0.214), while seed 456 is
 417 comparable (0.208). The +0.003 2p gain falls below the +0.010 pre-specified threshold, consistent
 418 with 2-hop providing incremental benefit primarily at longer depths.

419 G Inference Efficiency

420 Encoding—a one-time step including 2-hop edge adjacency construction—is 51.8× slower than
 421 GraphGPS. However, *per-query scoring* is 0.84–0.94× GraphGPS: DELTA is faster per query. In
 422 deployment scenarios where the graph is encoded once and queries are scored many times, DELTA’s
 423 amortized cost is comparable to or better than GraphGPS. Training cost (34.2×) is a one-time expense
 424 dominated by edge adjacency computation.

425 H Scaling Analysis

426 **Genuine but modest advantage at scale.** DELTA consistently outperforms DistMult at all con-
 427 trolled scales tested (N =500 to N =5,000), with the gap growing from +0.011 to +0.023. A fair
 428 comparison at N =14,541 requires matched training conditions and is left to future work.

Table 13: Phase 68 — Random-subgraph hop-depth ablation (3 seeds $\times$ 500 epochs, $N=2,242$ uniformly sampled entities, mean degree 7.6, hops=2 adjacency capped at 964K pairs). Values are mean $\pm$ std.

Config	LP MRR	1p MRR	2p MRR	3p MRR	2p $\rightarrow$ 3p
hops=1	0.229 $\pm$ 0.027	0.334 $\pm$ 0.040	0.242 $\pm$ 0.057	0.274 $\pm$ 0.060	+0.032
hops=2	0.233 $\pm$ 0.019	0.340 $\pm$ 0.017	0.249 $\pm$ 0.028	0.284 $\pm$ 0.031	+0.035
Δ (2-1)	+0.004	+0.006	+0.007	+0.010	—

Table 14: Inference timing comparison (mean of 10 runs, CUDA-synchronized).

Metric	DELTA-Matched	GraphGPS	Ratio
Encoding (once)	454.1 ms	8.8 ms	51.8 $\times$
1p per-query	777.7 μ s	921.9 μ s	0.84 $\times$
2p per-query	1,380 μ s	1,476 μ s	0.94 $\times$
3p per-query	1,251 μ s	1,371 μ s	0.91 $\times$
Training (total)	3,782 s	110 s	34.2 $\times$

Depth is the scaling bottleneck. At $N=2,000$, 3-layer DELTA catastrophically over-smooths (MRR 0.002, near random), while 1-layer DELTA surpasses DistMult. Residual gating eliminates the failure mode but gates freeze at $\sim 10\%$ layer / 90% residual, confirming that additional depth provides no benefit at this scale.

Top- k sparse attention. At $k=128$, sparse attention on 63M adjacency pairs matches full-softmax quality exactly (MRR 0.2472 vs. 0.2457), while $k=64$ degrades by -5.6% . Subsampling to $\sim 48\%$ retention at $N=5,000$ slightly improves MRR (+0.007), suggesting that a moderate number of high-quality neighbors outperforms access to all neighbors.

I Attention Temperature Asymmetry

We introduce learnable per-head temperature parameters $\tau^{(h)} = \exp(\theta^{(h)})$ that scale attention scores before softmax (Eq. 1 in the main paper). Higher τ produces sharper (more selective) attention; lower τ produces softer (more uniform) attention. We report these as observational findings; causal attribution requires controlled interventions (temperature freezing, per-head ablations) that we leave to future work.

Edge-node temperature divergence. Across all 20+ configurations tested, **edge attention temperatures drift upward** (from 4.0 to 4.4–5.0) while **node attention temperatures drift downward** (from 4.0 to 3.5–3.7). The model autonomously discovers that edges benefit from selective, discriminative attention while nodes benefit from broader, averaging attention.

Layer 0 is always dead. Regardless of initialization, first-layer attention heads remain near-uniform (entropy > 0.95). Initial entity embeddings carry no relational information worth selectively attending to; useful attention emerges only after the first reconciliation bridge provides cross-stream context. This suggests the bridge’s sequential cascade may initiate representational differentiation that later layers exploit—a causal question left for future bridge-ablation experiments.

Temperature controls operating mode. Three distinct operating regimes emerge:

1. **LP-optimized** (asymmetric: node=2, edge=6): LP MRR 0.491, sacrifices multi-hop.
2. **Balanced** (uniform temp=4.0): Best 3p MRR (0.402), moderate LP.
3. **Annealed** (node temp 4.0 $\rightarrow$ 2.0 during training): Breaks the 3p ceiling (0.415).

Temperature annealing creates compositional representations in early epochs that static initialization cannot replicate—training *trajectory* matters, not just final temperature. Multi-seed validation (3

Table 15: Scaling analysis: DELTA (1-layer) vs. DistMult across entity counts. All models on FB15k-237 subgraphs, seed=42. The full FB15k-237 row (Phase 67) uses top- $k=128$ sparse attention and 1,064K parameters; DM result is from published literature.

N	Triples	$ E_{\text{adj}} $	Test MRR		Δ	DELTA time	DM time	
			DELTA	DM				
500	9,700	256K	0.495	0.484	+0.011	3,782s	110s	[†] hops=1
2,000	62,700	15.2M	0.334	0.319	+0.015	5,788s	72s	
5,000	152,800	63.0M	0.247	0.224	+0.023	21,688s	1,784s	
14,541	310,116	29.8M [†]	0.205	0.241*	n/a*	—	—	

adjacency. ***Not directly comparable:** the published DistMult result [2] uses tuned hyperparameters and extended training time; our DELTA result at $N=14,541$ uses a fixed training budget. The $\Delta=-0.036$ entry does *not* establish that DELTA underperforms DistMult under matched conditions at full scale.

seeds) confirms LP improvements are statistically robust (MRR 0.484 ± 0.005); multi-hop temperature effects showed high cross-seed variance that resolved with larger query counts (10,000 samples, reducing std by 66–84%).

J Detailed Limitations

Scale and benchmark coverage (critical). Compositional reasoning experiments use a 500-entity degree-biased subgraph (mean degree ≈ 19.7), far denser than the full graph (≈ 4.1). The strong multi-hop results (0.758–0.790 MRR, 5-hop) are demonstrated on this small subgraph. On the complete 14,541-entity FB15k-237 graph, DELTA achieves 1p MRR ≈ 0.205 , substantially below published SOTA: RotatE 0.338, CompGCN 0.355, NBFNet 0.415. Evaluation on other standard benchmarks (YAGO3-10, WN18RR [24], ogbl-wikikg2) is an important open direction.

Baseline coverage (important). We compare primarily against general graph transformers (GraphGPS, GRIT) and DistMult. Specialized path-based KG reasoners (NBFNet [8], A*Net [21]) are absent. Our multi-hop evaluation uses internally generated chain queries with soft entity traversal—not the established BetaE/ConE/GQE complex-query benchmarks, which use different query generation protocols and logical operators. Results are not directly comparable to published multi-hop reasoning numbers from those systems.

Full-scale multi-hop evaluation (important). The Phase 67 full-graph ablation evaluates hops=1 vs. hops=2 but does not include the full 2-hop→5-hop depth sweep from the subgraph experiments. Whether DELTA’s depth-monotonic improvement persists on the full sparse graph has not been directly validated.

Computational cost (notable). The one-time encoding overhead ($51.8 \times$ GraphGPS) limits applicability to static graphs. The full-graph (Phase 67) experiment required approximately 3 days on an NVIDIA H200 GPU with 141 GB VRAM to evaluate 6 conditions (3 seeds $\times$ 2 hop settings $\times$ 200 epochs), with seeds running in parallel.

Causal claims (methodological). The attention temperature asymmetry is a consistent observational pattern, not a causally established property. Causal attribution requires controlled interventions (temperature freezing, head-type ablations) not performed here. BrainEncoder (Appendix C) and domain transfer (Appendix D) results are preliminary signals from small experiments and should not be treated as validated findings.

Subgraph selection bias (methodological). Using top-degree entities for primary experiments biases toward entities with rich relational context and a highly saturated edge adjacency graph, which may affect which architectural choices appear beneficial relative to sparser realistic settings.

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Justification: The abstract and introduction claim (1) that edge-to-edge attention over multi-hop adjacency improves compositional reasoning on knowledge graphs, and (2) that DELTA demonstrates consistent improvements on synthetic, full-graph, and subgraph benchmarks. All claims are directly supported by the experimental results in §4 with numerical evidence.

2. Limitations

Question: Does the paper discuss the limitations of the work?

Answer: [Yes]

Justification: Limitations are addressed in §5 (summary) and Appendix J (detailed): the gap between SOTA and DELTA’s full-graph LP; the Phase 66 dense-subgraph ablation finding all conditions statistically indistinguishable; the preliminary nature of BrainEncoder and domain transfer results; and missing baselines (NBFNet, A*Net).

3. Theory Assumptions and Proofs

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4. Experimental Reproducibility Details

Question: Does the paper fully describe the experimental setup to enable re-implementation?

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Justification: §4 specifies dataset (FB15k-237, publicly available), subgraph construction procedure (top- k degree filter), model hyperparameters (hidden dimensions, number of layers, hop depth, learning rate, epochs, DropEdge rate), evaluation metrics, and the number of seeds for each experiment. Phase identifiers map to distinct experimental conditions described in the text. Code covering the edge adjacency construction, top- k sparse attention implementation, and multi-hop query generation procedure will be released upon acceptance.

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